Exercises

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Exercise 1

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   i. Embed the input sequence

Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   ii. Apply positional encoding

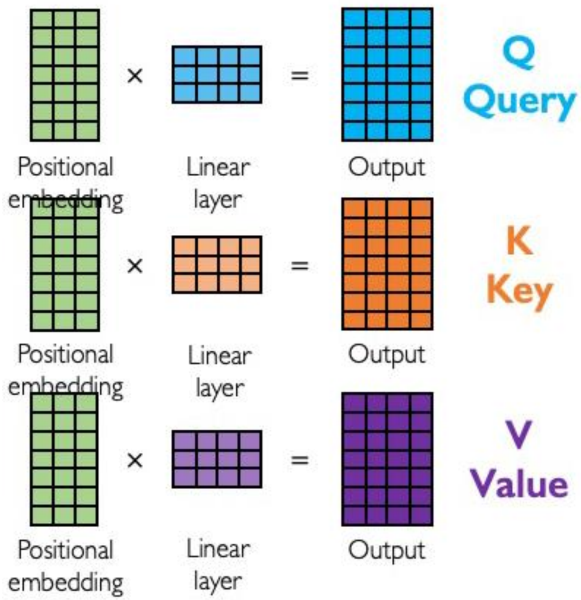
$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   iii. Compute the query, key, and value matrices (**Q**, **K**, **V**)



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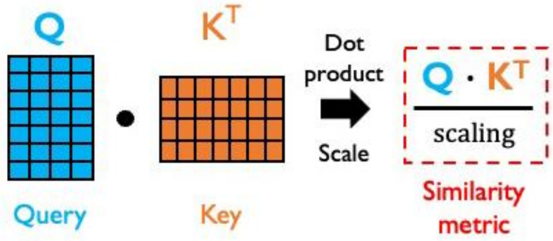
Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

    iv. Compute the (unnormalized) attention scores (**S**) (e.g., by using the scaled dot-product attention)

$$S = \frac{Q \cdot K^T}{\text{scaling}}$$



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where $\text{scaling} = \sqrt{d_k}$, with $d_k = dim(k)$

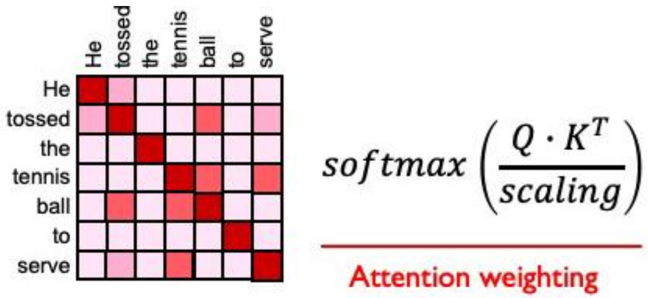
Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   v. Compute the (normalized) attention weights (**P**)

$$P = \mathrm{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right)$$



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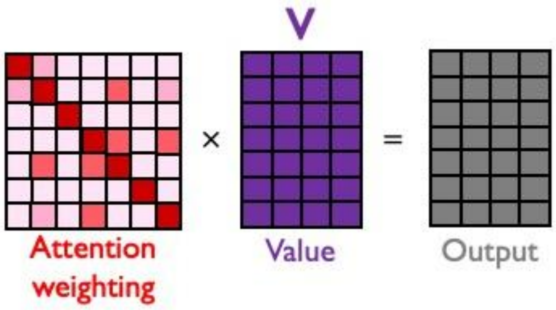
Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   vi. Compute the (single-head$_i$) output matrix ($\mathbf{Z_i}$)

$$Z_i = P \cdot V = \text{Attention}\left(Q_i, K_i, V_i\right)$$



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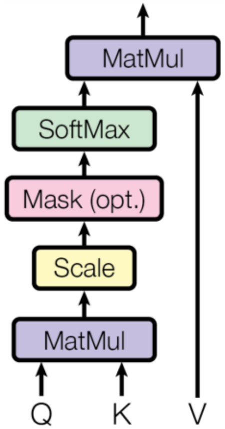


Vaswani et al. (NIPS 2017)

Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   vii. Repeat steps iii. to vi. for the other h-1 attention heads (with different projection parameter matrices)

$$Z_1 = \text{Attention}\left(Q_1, K_1, V_1\right)$$
$$Z_2 = \text{Attention}\left(Q_2, K_2, V_2\right)$$
$$\cdots$$
$$Z_h = \text{Attention}\left(Q_h, K_h, V_h\right)$$

Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   viii. Concatenate the output matrices from the h attention heads

$$\mathrm{Concat}\left(Z_1, Z_2, \ldots, Z_h\right)$$

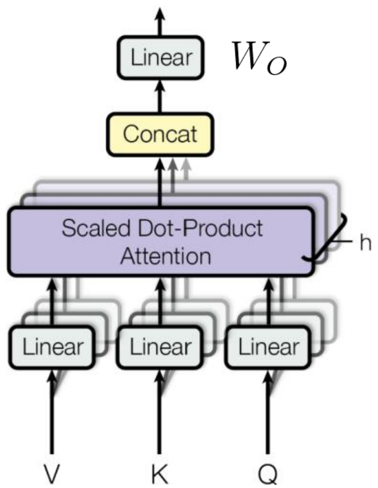
Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

    ix. Compute the (multi-head) output matrix ($\mathbf{Z_{tot}}$)

$$Z_{tot} = \mathrm{Concat}\left(Z_1, \ldots, Z_h\right) \cdot W_O = \mathrm{MultiHeadAttention}\left(Q, K, V\right)$$



Vaswani et al. (NIPS 2017)

Exercise 1 – Solution

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   i. Embed the input sequence
   ii. Apply positional encoding
   iii. Compute the query, key, and value matrices ($\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$)
   iv. Compute the (unnormalized) attention scores ($\mathbf{S}$) (e.g., by using the scaled dot-product attention)
   v. Compute the (normalized) attention weights ($\mathbf{P}$)
   vi. Compute the (single-head$_i$) output matrix ($\mathbf{Z_i}$)
   vii. Repeat steps iii. to vi. for the other h-1 attention heads (with different projection parameter matrices)
   viii. Concatenate the output matrices from the h attention heads
   ix. Compute the (multi-head) output matrix ($\mathbf{Z_{tot}}$)

Exercise 1

Considering the attention mechanism:

1. Enumerate its steps (providing the mathematical formulation whenever possible)

   Okay but... practically speaking?

Exercise 1

Considering the attention mechanism:

2. Let $\mathbf{x}_1 = [-1.0, 2.0, 0.5]$ $\mathbf{x}_2 = [-0.5, 2.0, -1.5]$ $\mathbf{x}_3 = [-1.0, 1.5, -0.5]$ $\mathbf{x}_4 = [-1.5, 1.5, 0.5]$

   be a sequence of length 4, where each element is a vector $\in \mathbb{R}^3$. Assume that positional encoding has already been applied.

   Suppose that we want to compute the (single-head) self-attention on this input. Assuming that the projection matrices for queries, keys, and values are respectively:

$$\mathbf{W}_Q = \begin{bmatrix} -1.0 & 1.5 \\ 2.5 & 0.5 \\ -0.5 & -1.5 \end{bmatrix} \qquad \mathbf{W}_K = \begin{bmatrix} -1.5 & -0.5 \\ 2.0 & 1.5 \\ -0.5 & -2.5 \end{bmatrix} \qquad \mathbf{W}_V = \begin{bmatrix} -1.5 & -0.5 \\ -0.5 & -2.0 \\ 0.5 & 0.0 \end{bmatrix}$$

   Compute the query, key, and value matrices (**Q**, **K**, **V**) and the resulting output matrix (**Z**).

Exercise 1 – Solution

We have the input matrix (assuming already position-encoded):

$$\mathbf{X} = \begin{bmatrix} -1.0 & 2.0 & 0.5 \\ -0.5 & 2.0 & -1.5 \\ -1.0 & 1.5 & -0.5 \\ -1.5 & 1.5 & 0.5 \end{bmatrix}$$

We need to compute the query, key, and value matrices which are defined as follows:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q \qquad \mathbf{K} = \mathbf{X}\mathbf{W}_K \qquad \mathbf{V} = \mathbf{X}\mathbf{W}_V$$

Exercise 1 – Solution

Let's compute the query matrix $\mathbf{Q}$:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q = \begin{bmatrix} -1.0 & 2.0 & 0.5 \\ -0.5 & 2.0 & -1.5 \\ -1.0 & 1.5 & -0.5 \\ -1.5 & 1.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} -1.0 & 1.5 \\ 2.5 & 0.5 \\ -0.5 & -1.5 \end{bmatrix} = \begin{bmatrix} \cdots & \cdots \\ \cdots & \cdots \\ \cdots & \cdots \\ \cdots & \cdots \end{bmatrix}$$

Exercise 1 – Solution

Let's compute the query matrix $\mathbf{Q}$:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q = \begin{bmatrix} -1.0 & 2.0 & 0.5 \\ -0.5 & 2.0 & -1.5 \\ -1.0 & 1.5 & -0.5 \\ -1.5 & 1.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} -1.0 & 1.5 \\ 2.5 & 0.5 \\ -0.5 & -1.5 \end{bmatrix} = \begin{bmatrix} 5.75 & -1.25 \\ 6.25 & 2.50 \\ 5.00 & 0.00 \\ 5.00 & -2.25 \end{bmatrix}$$

Exercise 1 – Solution

Let's compute the key matrix **K**:

$$\mathbf{K} = \mathbf{X}\mathbf{W}_K = \begin{bmatrix} -1.0 & 2.0 & 0.5 \\ -0.5 & 2.0 & -1.5 \\ -1.0 & 1.5 & -0.5 \\ -1.5 & 1.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} -1.5 & -0.5 \\ 2.0 & 1.5 \\ -0.5 & -2.5 \end{bmatrix} = \begin{bmatrix} \dots & \dots \\ \dots & \dots \\ \dots & \dots \\ \dots & \dots \end{bmatrix}$$

Exercise 1 – Solution

Let's compute the key matrix **K**:

$$\mathbf{K} = \mathbf{X}\mathbf{W}_K = \begin{bmatrix} -1.0 & 2.0 & 0.5 \\ -0.5 & 2.0 & -1.5 \\ -1.0 & 1.5 & -0.5 \\ -1.5 & 1.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} -1.5 & -0.5 \\ 2.0 & 1.5 \\ -0.5 & -2.5 \end{bmatrix} = \begin{bmatrix} 5.25 & 2.25 \\ 5.50 & 7.00 \\ 4.75 & 4.00 \\ 5.00 & 1.75 \end{bmatrix}$$

Exercise 1 – Solution

Let's compute the value matrix **V**:

$$\mathbf{V} = \mathbf{X}\mathbf{W}_V = \begin{bmatrix} -1.0 & 2.0 & 0.5 \\ -0.5 & 2.0 & -1.5 \\ -1.0 & 1.5 & -0.5 \\ -1.5 & 1.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} -1.5 & -0.5 \\ -0.5 & -2.0 \\ 0.5 & 0.0 \end{bmatrix} = \begin{bmatrix} \dots & \dots \\ \dots & \dots \\ \dots & \dots \\ \dots & \dots \end{bmatrix}$$

Exercise 1 – Solution

Let's compute the value matrix **V**:

$$\mathbf{V} = \mathbf{X}\mathbf{W}_V = \begin{bmatrix} -1.0 & 2.0 & 0.5 \\ -0.5 & 2.0 & -1.5 \\ -1.0 & 1.5 & -0.5 \\ -1.5 & 1.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} -1.5 & -0.5 \\ -0.5 & -2.0 \\ 0.5 & 0.0 \end{bmatrix} = \begin{bmatrix} 0.75 & -3.50 \\ -1.00 & -3.75 \\ 0.50 & -2.50 \\ 1.75 & -2.25 \end{bmatrix}$$

Exercise 1 – Solution

Query, key, and value matrices:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q = \begin{bmatrix} 5.75 & -1.25 \\ 6.25 & 2.50 \\ 5.00 & 0.00 \\ 5.00 & -2.25 \end{bmatrix}$$

$$\mathbf{K} = \mathbf{X}\mathbf{W}_K = \begin{bmatrix} 5.25 & 2.25 \\ 5.50 & 7.00 \\ 4.75 & 4.00 \\ 5.00 & 1.75 \end{bmatrix}$$

$$\mathbf{V} = \mathbf{X}\mathbf{W}_V = \begin{bmatrix} 0.75 & -3.50 \\ -1.00 & -3.75 \\ 0.50 & -2.50 \\ 1.75 & -2.25 \end{bmatrix}$$

Exercise 1 – Solution

Now we need to compute the (unnormalized) attention scores (**S**), e.g. by using the scaled dot-product attention.

$$S = \frac{Q \cdot K^T}{\text{scaling}} \qquad \text{scaling} = \sqrt{d_k} = \sqrt{dim(k)} = \sqrt{2}$$

$$\mathbf{S} = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 5.75 & -1.25 \\ 6.25 & 2.50 \\ 5.00 & 0.00 \\ 5.00 & -2.25 \end{bmatrix} \cdot \begin{bmatrix} 5.25 & 5.50 & 4.75 & 5.00 \\ 2.25 & 7.00 & 4.00 & 1.75 \end{bmatrix} = \begin{bmatrix} \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \end{bmatrix}$$

Exercise 1 – Solution

Now we need to compute the (unnormalized) attention scores (**S**), e.g. by using the scaled dot-product attention.

$$S = \frac{Q \cdot K^T}{\text{scaling}} \qquad \text{scaling} = \sqrt{d_k} = \sqrt{dim(k)} = \sqrt{2}$$

$$\mathbf{S} = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 5.75 & -1.25 \\ 6.25 & 2.50 \\ 5.00 & 0.00 \\ 5.00 & -2.25 \end{bmatrix} \cdot \begin{bmatrix} 5.25 & 5.50 & 4.75 & 5.00 \\ 2.25 & 7.00 & 4.00 & 1.75 \end{bmatrix} = \begin{bmatrix} 19.36 & 16.17 & 15.78 & 18.78 \\ 27.18 & 36.68 & 28.06 & 25.19 \\ 18.56 & 19.44 & 16.79 & 17.68 \\ 14.98 & 8.31 & 10.43 & 14.89 \end{bmatrix}$$

Exercise 1 – Solution

Now we can compute the (normalized) attention weights (**P**) by applying softmax normalization <u>row-wise</u>

$$P = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right)$$

$$\mathbf{P} = \text{softmax}\left(\begin{bmatrix} 19.36 & 16.17 & 15.78 & 18.78 \\ 27.18 & 36.68 & 28.06 & 25.19 \\ 18.56 & 19.44 & 16.79 & 17.68 \\ 14.98 & 8.31 & 10.43 & 14.89 \end{bmatrix}\right) = \begin{bmatrix} ... & ... & ... & ... \\ ... & ... & ... & ... \\ ... & ... & ... & ... \\ ... & ... & ... & ... \end{bmatrix}$$

Exercise 1 – Solution

Now we can compute the (normalized) attention weights (**P**) by applying softmax normalization <u>row-wise</u>

$$P = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right)$$

$$\mathbf{P} = \text{softmax}\left(\begin{bmatrix} 19.36 & 16.17 & 15.78 & 18.78 \\ 27.18 & 36.68 & 28.06 & 25.19 \\ 18.56 & 19.44 & 16.79 & 17.68 \\ 14.98 & 8.31 & 10.43 & 14.89 \end{bmatrix}\right) = \begin{bmatrix} 6.13\text{e-}01 & 2.54\text{e-}02 & 1.71\text{e-}02 & 3.45\text{e-}01 \\ 7.47\text{e-}05 & 9.99\text{e-}01 & 1.81\text{e-}04 & 1.02\text{e-}05 \\ 2.49\text{e-}01 & 6.04\text{e-}01 & 4.26\text{e-}02 & 1.03\text{e-}01 \\ 5.19\text{e-}01 & 6.56\text{e-}04 & 5.47\text{e-}03 & 4.75\text{e-}01 \end{bmatrix}$$

Exercise 1 – Solution

Finally, the resulting output matrix (**Z**) is given by:

$$Z = P \cdot V = \text{Attention}\left(Q, K, V\right)$$

$$\mathbf{Z} = \begin{bmatrix} 6.13\text{e-}01 & 2.54\text{e-}02 & 1.71\text{e-}02 & 3.45\text{e-}01 \\ 7.47\text{e-}05 & 9.99\text{e-}01 & 1.81\text{e-}04 & 1.02\text{e-}05 \\ 2.49\text{e-}01 & 6.04\text{e-}01 & 4.26\text{e-}02 & 1.03\text{e-}01 \\ 5.19\text{e-}01 & 6.56\text{e-}04 & 5.47\text{e-}03 & 4.75\text{e-}01 \end{bmatrix} \cdot \begin{bmatrix} 0.75 & -3.50 \\ -1.00 & -3.75 \\ 0.50 & -2.50 \\ 1.75 & -2.25 \end{bmatrix} = \begin{bmatrix} \dots & \dots \\ \dots & \dots \\ \dots & \dots \\ \dots & \dots \end{bmatrix}$$

Exercise 1 – Solution

Finally, the resulting output matrix (**Z**) is given by:

$$Z = P \cdot V = \text{Attention}\left(Q, K, V\right)$$

$$\mathbf{Z} = \begin{bmatrix} 6.13\text{e-}01 & 2.54\text{e-}02 & 1.71\text{e-}02 & 3.45\text{e-}01 \\ 7.47\text{e-}05 & 9.99\text{e-}01 & 1.81\text{e-}04 & 1.02\text{e-}05 \\ 2.49\text{e-}01 & 6.04\text{e-}01 & 4.26\text{e-}02 & 1.03\text{e-}01 \\ 5.19\text{e-}01 & 6.56\text{e-}04 & 5.47\text{e-}03 & 4.75\text{e-}01 \end{bmatrix} \cdot \begin{bmatrix} 0.75 & -3.50 \\ -1.00 & -3.75 \\ 0.50 & -2.50 \\ 1.75 & -2.25 \end{bmatrix} = \begin{bmatrix} 1.05 & -3.06 \\ -1.00 & -3.75 \\ -0.21 & -3.48 \\ 1.22 & -2.90 \end{bmatrix}$$

Exercise 1

Considering the attention mechanism:

3. Write the corresponding Python code.

Exercise 1 – Solution

Considering the attention mechanism:

3. Write the corresponding Python code.

```python
import numpy as np
from scipy.special import softmax

x_1 = [-1.0, 2.0,  0.5]
x_2 = [-0.5, 2.0, -1.5]
x_3 = [-1.0, 1.5, -0.5]
x_4 = [-1.5, 1.5,  0.5]
X = np.stack([x_1, x_2, x_3, x_4], axis=0)

W_Q = np.array([[-1.0,  1.5],
                [ 2.5,  0.5],
                [-0.5, -1.5]])

W_K = np.array([[-1.5, -0.5],
                [ 2.0,  1.5],
                [-0.5, -2.5]])

W_V = np.array([[-1.5, -0.5],
                [-0.5, -2.0],
                [ 0.5,  0.0]])

Q = X @ W_Q
K = X @ W_K
V = X @ W_V

attention_scores = Q @ K.transpose()
d_k = K.shape[1]

attention_scores /= np.sqrt(d_k)

attention_weights = softmax(attention_scores, axis=1)

Z = attention_weights @ V
```


Exercise 1

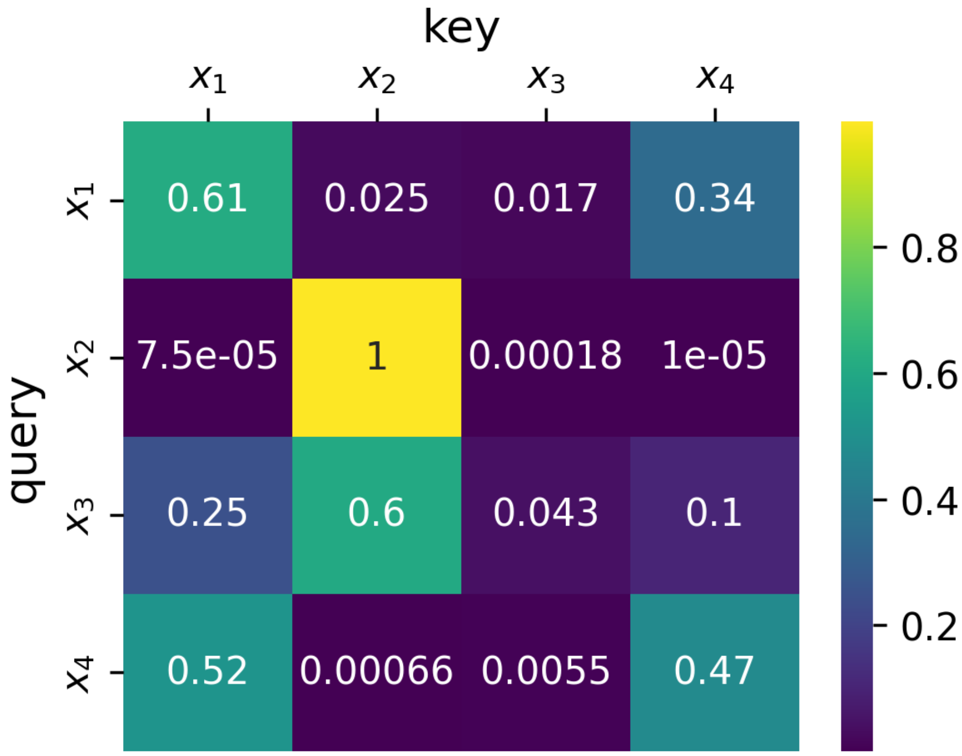
Considering the attention mechanism:

4. Plot the attention map.

Exercise 1 – Solution

Considering the attention mechanism:

4. Plot the attention map.



Acknowledgements and copyright license

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  - I would like to thank Dr. Giuseppe Gallipoli, who collaborated to the revision of the teaching content.

- Affiliation
  - The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
    - https://dbdmg.polito.it
    - https://smartdata.polito.it

Thank you!